Fiscal Year:	FY 2017	**Task Last Updated:**	FY 12/28/2017
PI Name:	Buras, William M.Sc.		
Project Title:	Novel Methods for the Flexible Ultrasound System Utilizing Augmented Reality Just-In-Time Procedural Guidance		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **ExMC**:Exploration Medical Capabilities		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Houston	**State:**	TX
Zip Code:	77058-2711	**Congressional District:**	36
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	SBIR Phase II
Start Date:	05/31/2017	**End Date:**	05/30/2019
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
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Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:			
Performance Goal No.:			
Performance Goal Text:			
	NASA's future manned spaceflight missions will require medical diagnosis and treatment capabilities that address both the anticipated health risks and perform well in austere, remote operational environments. Spaceflight- ready medical devices will need to be capable of an increased degree of autonomous operation, acquiring clinically relevant and diagnosable data by every astronaut, not just select physician crew members credentialed in spaceflight medicine. Ultrasound is a diagnostic and treatment technology that currently fulfills mission medical capability support on the International Space Station (ISS) and is planned to accompany future deep-space missions. The Flexible Ultrasound System (FUS) is a new platform that is currently being developed by NASA and research partners to support this mission role. We propose three specific aims for this project proposal for methodological development utilizing the FUS platform: 1.) Define a list of highest priority/yield ultrasound methods to develop and implement using the FUS and our guidance prototype through careful review of the HRP (Human Research Program) roadmap and direct discussion with ExMC (Exploration Medical Capability) Physicians. 2.) Expand an Augmented Reality (AR) user interface for these		

Task Description:	ultrasound methods that provides procedural guidance in acquiring and initially diagnosing sonographic data for ultrasound procedures to enhanced degree of procedural competency. 3.) Develop and test the integration of magnetic-based Volume Navigation, and complementary dimensional referencing technologies, on the FUS platform to allow for 3-dimensional ultrasound procedural guidance through the Head Mounted Display. 4.) Integrate the current system with procedural guidance resources such as Electronic Procedures (eProc) and existing ultrasound training materials. 5.) Develop deep machine learning capabilities on the FUS platform to enable image recognition and analysis that can aid in not only quality image acquisition, but also analysis in austere, and remote operational environments. POTENTIAL NASA COMMERCIAL APPLICATIONS: This project maintains the spirit of the goals emphasized in the NASA strategy to revitalize and expand our investments in technology, commercial spaceflight, and robotic exploration precursors. NASA's multi-destination human space exploration strategy as well as its ambitious program of innovative robotics missions will challenge engineers to develop these new and complex systems with advanced capabilities. The agency is exploring multiple destinations. It plans to conduct increasingly complex missions to a range of destinations beyond low Earth orbit (LEO), including cis-lunar space, near-Earth asteroids (NEAs), the moon, and Mars and its moons. VULCAN will be one of the medical tools for the Journey to Mars in the 2030s.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	POTENTIAL NON-NASA COMMERCIAL APPLICATIONS: Tietronix has already initiated work with Methodist Houston and an NSF (National Science Foundation) sponsored Cyber-systems of the future Operating Room that is an academic/industry consortium (with membership such as Medtronic, Boston Scientific, and Karl Storz) on developing this technology for terrestrial medicine.
Task Progress:	New project for FY2017. Reporting not required for this SBIR Phase 2 project.
Bibliography Type:	Description: (Last Updated:)